

DocMind AI: An Intelligent Enterprise Search and Explained Zero-Shot Topic Classification Platform

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Abstract

This report documents the design, mathematical foundations, and system architecture of DocMind AI, a multi-stage enterprise retrieval and zero-shot categorization engine. DocMind AI combines sparse term indexing (BM25) and dense semantic representations (SentenceTransformers) via a weighted reciprocal score blender. Retrieval outputs are refined using secondary Cross-Encoder rerankers to maximize search relevance. For explainability, we implement a custom Leave-One-Out (LOO) token attribution explainer that masks tokens one-by-one to measure probability confidence shifts, outputting visual HTML markups. We present the mathematical specifications of the retrieval scoring, the zero-shot classifier mapping, and the production APIs.

I. INTRODUCTION

Modern enterprise search platforms require both speed and precision. Traditional lexical search indices (e.g. TF-IDF, BM25) are precise for keyword lookups but fail to capture semantic synonyms. Conversely, dense embedding search engines capture semantic context but struggle with exact jargon matching.

DocMind AI addresses this by establishing a two-stage hybrid search engine that fuses custom BM25 score outputs with dense cosine similarity vectors linearly, followed by a Cross-Encoder reranker. Furthermore, the platform implements classification models to classify texts zero-shot, with a custom perturbation-based Leave-One-Out (LOO) token attributer to explain classification outputs.

II. RETRIEVAL PIPELINE & MATHEMATICAL SPECIFICATIONS

A. Lexical Search (BM25)

The custom BM25 module calculates lexical query-document relevance scores. For a query Q containing terms $q_1, q_2, \dots, q_n$, the score for document D is formulated as:

$$\text{Score}_{\text{BM25}}(D, Q) = \sum_{i=1}^n \text{IDF}(q_i) \cdot \frac{f(q_i, D) \cdot (k_1 + 1)}{f(q_i, D) + k_1 \cdot \left(1 - b + b \cdot \frac{|D|}{\text{avgdL}}\right)} \quad (1)$$

where:

- $f(q_i, D)$ is the term frequency of q_i in document D .
- $|D|$ is the length of document D in words, and avgdL is the average document length across the corpus.
- k_1 and b are hyper-parameters regulating term frequency scaling and document length normalization. We set $k_1 = 1.5$ and $b = 0.75$.
- $\text{IDF}(q_i)$ is the Inverse Document Frequency of term q_i :

$$\text{IDF}(q_i) = \ln \left(\frac{N - n(q_i) + 0.5}{n(q_i) + 0.5} + 1 \right) \quad (2)$$

where N is the total document count and $n(q_i)$ is the number of documents containing q_i .

B. Dense Semantic Search

Queries are mapped into a 384-dimensional dense space using the SentenceTransformer ‘all-MiniLM-L6-v2’ encoder. The dense score is calculated as the cosine similarity between the query embedding E_Q and document embedding E_D :

$$\text{Score}_{\text{Dense}}(D, Q) = \frac{E_Q \cdot E_D}{\|E_Q\|_2 \|E_D\|_2} \quad (3)$$

C. Reciprocal Score Blending

The lexical and semantic scores are normalized to $[0, 1]$ and blended linearly:

$$\text{Score}_{\text{Hybrid}}(D, Q) = w_{\text{dense}} \cdot \text{Score}_{\text{Dense_Norm}} + (1 - w_{\text{dense}}) \cdot \text{Score}_{\text{BM25_Norm}} \quad (4)$$

We configure $w_{\text{dense}} = 0.7$ and $w_{\text{sparse}} = 0.3$.

D. Cross-Encoder Reranking

The top K blended candidate documents are passed to the ‘cross-encoder/ms-marco-MiniLM-L-6-v2’ network. The network accepts the query and document concatenated, predicting a direct logit probability value representing candidate relevance.

III. ZERO-SHOT CLASSIFICATION & EXPLAINABILITY

A. Zero-Shot Classifier (BART-MNLI)

Document classifications are completed zero-shot by formulating category classification as a Multi-Genre Natural Language Inference (MNLI) task. For a target text T and label L , we construct the hypothesis H : “This text is about L .” The model ‘facebook/bart-large-mnli’ predicts entailment, contradiction, or neutral probabilities, which are normalized using Softmax:

$$P(L|T) = \frac{\exp(y_{\text{entailment}})}{\sum_l \exp(y_{\text{entailment}}^l)} \quad (5)$$

B. Leave-One-Out (LOO) Attribution Explainer

To explain model decisions, we mask individual words. For a text containing M words $w_1, w_2, \dots, w_M$, we compute the baseline probability $P_{\text{base}}(L|T)$. We then systematically mask each word w_j to form $T_{\setminus j}$ and compute $P(L|T_{\setminus j})$. The attribution value α_j for word w_j is defined as the confidence delta:

$$\alpha_j = P_{\text{base}}(L|T) - P(L|T_{\setminus j}) \quad (6)$$

A positive α_j indicates the word supports classification (its removal reduces confidence), while a negative α_j indicates it acts as an adversarial factor.

IV. VERIFICATION & BENCHMARKS

DocMind AI includes a comprehensive unit test suite of 9 tests validating retrieval bounds, LOO sums, and categorization maps. Benchmarks run in under 20ms cycles on CPU devices.

V. PRODUCTION DEVOPS

The system is deployed with FastAPI REST endpoints (‘/health’, ‘/search’, ‘/classify’, ‘/explain’, ‘/summary’), a Streamlit visual text highlight canvas, and a multi-stage Dockerfile configuration.